



Koneru Lakshmaiah Education Foundation

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SECTION : DBSE&DBD Y25 Section-8

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week -3

Scenario: BookFlow wants to capture user reviews and technical specs for books, where every book has different details. A fixed SQL table is too restrictive, so MongoDB is used to manage this unstructured content and generate a "Top Rated" report using the aggregation pipeline.

Tasks: 1) Document Modeling — create a book_metadata collection with embedded patterns (reviews list inside the book document) and referenced patterns (linking to member_id in SQL). 2) Advanced Querying — use comparison operators (\$gt, \$in) to find books rated above 4, and a regex query to find books with "Digital" formats. 3) Aggregation Pipeline — match books published after 2020, \$unwind the reviews array, then \$group by title and compute the average rating. 4) Indexing for Speed — create a text index on the review comments field and use .explain() to check whether the query uses the index.

Skills gained: BSON modelling and embed-vs-reference decisions, pipeline engineering with \$match/\$group/\$unwind, multikey/text index performance tuning, and handling polymorphic documents.

Task 1: Document Modelling

The problem with a fixed SQL table

In a relational model, every book would need a row in a books table with a fixed set of columns: pages, file_size, narrator, runtime, binding_type, drm_flag, and so on. Since a hardcover doesn't have a narrator and an audiobook doesn't have page count, most of these columns would be NULL for any given row. As BookFlow adds new formats (for example, a future "interactive ebook" with different specs again), the table needs a schema migration (ALTER TABLE) every time. This is the classic impedance mismatch between rigid relational schemas and variable, semi-structured content.

Why MongoDB fits

MongoDB stores data as BSON documents (a binary superset of JSON) inside collections, and documents in the same collection are not required to share the same fields. This is called a flexible or polymorphic schema. Each book document only carries the fields relevant to its own format — no wasted NULL columns, no migrations needed to add a new book type.

The core modelling decision: embed vs. reference

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Project 0

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Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
- book_metadata**
- local
- sample_mflix

book_metadata

+

Cluster0 > bookflow > book_metadata

View monitoring

Visualize Your Data

Documents 5

Aggregations

Schema

Indexes 1

Validation

```
{  "reviews.rating": {    "$gt": 4  }}
```

Explain

Reset

Find

Options

INSIGHT

25 1 - 5 of 5

```
_id: ObjectId('6a7e871365238b575c2e2ee0')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)
```

Alert System Status: All Good

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```
{  "reviews.rating": {    "$gt": 4  }}
```

Explain

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```
reviews: Array (3)
```

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Indexes 1

Validation

```
{  "reviews.rating": {    "$in": [4, 5]  }}
```

Explain

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Options

INSIGHT

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```
_id: ObjectId('6a7e871365238b575c2e2ee0')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)
```

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book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

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Explain Reset Find Options

```
{
  "format": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

25 1 - 4 of 4

```
{
  "_id": ObjectId('6a7e871365230b575c2e2ee0')
  book_id: 101
  title: "The Martian"
  author: "Andy Weir"
  published_year: 2021
  category: "Science Fiction"
  format: "Digital PDF"
  file_size: "2.5 MB"
  member_id: 101
}
```

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Aggregation Pipeline

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book_metadata

Cluster0 > bookflow > book_metadata

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View monitoring Visualize Your Data

Generate aggregation Explain Run Options

Untitled - modified SAVE CREATE NEW EXPORT CODE PREVIEW STAGES TEXT WIZARD

Stage 1 \$match

```
1 {
2   "published_year": {
3     "$gt": 2020
4   }
5 }
```

Output preview after \$match stage (Sample of 4 documents)

```
{
  "_id": ObjectId('6a7e871365230b575c2e2ee0')
  book_id: 101
  title: "The Martian"
  author: "Andy Weir"
  published_year: 2021
  category: "Science Fiction"
  format: "Digital PDF"
  file_size: "2.5 MB"
}
```

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CLUSTERS (1)

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Search clusters

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Cluster0

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admin

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bookflow

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book_meta...

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local

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sample_mflix

book_metadata

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Cluster0 > bookflow > book_metadata

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View monitoring

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Visualize Your Data

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Documents 5

Aggregations

Schema

Indexes 1

Validation

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\$match

\$unwind

Generate aggregation

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Explain

Run

Options ▶

Untitled - modified

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SAVE

+ CREATE NEW

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EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

⚙️

▼

Stage 2

\$unwind

🔍

1

2

3

{

"path": "\$reviews"

}

Output preview after \$unwind stage (Sample of 10 documents)

_id: ObjectId('6a7e871365230b575c2e2ee0')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

_id: Obj

book_id:

title: '

author:

publishe

category

format:

file_siz

member i

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CLUSTERS (1)

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Search clusters

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Cluster0

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admin

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bookflow

▶

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▶

local

▶

sample_mflix

book_metadata

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Cluster0 > bookflow > book_metadata

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\$match

\$unwind

\$group

Generate aggregation

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🔍

Explain

Run

Options ▶

Untitled - modified

📁

SAVE

+ CREATE NEW

📄

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

⚙️

▼

Stage 3

\$group

🔍

1

2

3

4

5

6

7

8

9

{

"_id": "\$title",

"average_rating": {

"\$avg": "\$reviews.rating"

},

"total_reviews": {

"\$sum": 1

}

}

Output preview after \$group stage (Sample of 4 documents)

_id: "The Martian"

average_rating: 4.666666666666667

total_reviews: 3

_id: "Da

average_

total_re

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▶️ admin

▶️ bookflow

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book_metadata

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Cluster0 > bookflow > book_metadata

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\$match

\$unwind

\$group

\$sort

Generate aggregation

⚙️

Explain

Run

Options ▶

Untitled - modified

💾 SAVE

➕ CREATE NEW

📄 EXPORT CODE

PREVIEW

STAGES

📄 TEXT

WIZARD

⚙️

▼

stage 1

🔍

🔍

1 {

2 "average_rating": -1

3 }

Output preview after \$sort stage (Sample of 4 documents)

_id: "The Martian"

average_rating : 4.666666666666667

total_reviews : 3

_id: "Du"

average_re

total_re

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Cluster0

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Validation

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\$match

\$unwind

\$group

\$sort

Edit

⚙️

Explain

Run

Options ▶

ALL RESULTS

📄 EXPORT CODE

Showing 1 - 4 count results

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⌵

⌵

⌵

_id: "Dune"

average_rating : 4.666666666666667

total_reviews : 3

_id: "The Martian"

average_rating : 4.666666666666667

total_reviews : 3

_id: "Artificial Intelligence Basics"

average_rating : 4.333333333333333

total_reviews : 3

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book_metadata

Cluster0 > bookflow > book_metadata

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\$match \$unwind \$group \$sort Edit Explain Run Options

ALL RESULTS EXPORT CODE Showing 1 - 4 count results

```
average_rating : 4.666666666666667
total_reviews : 3

_id: "Artificial Intelligence Basics"
average_rating : 4.333333333333333
total_reviews : 3

_id: "Data Structures"
average_rating : 4
total_reviews : 2
```

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Text Index

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Documents 5 Aggregations Schema **Indexes 2** Validation

Create Index Refresh

Looking for search indexes?
These indexes can be created and viewed under [Search and Vector Search](#).

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	36.9 kB	8 (since Fri Aug 14 2026)	UNIQUE	READY
> reviews.comments_text	TEXT	20.5 kB	0 (since Fri Aug 14 2026)		READY

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<https://docs.mongodb.org/manual/core/index-text/>

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Indexes 2

Validation

```
{
  "$text": {
    "$search": "excellent"
  }
}
```

Generate query

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

1 - 5 of 5

```
_id: ObjectId('6a7e871365238b575c2e2ee0')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)
```

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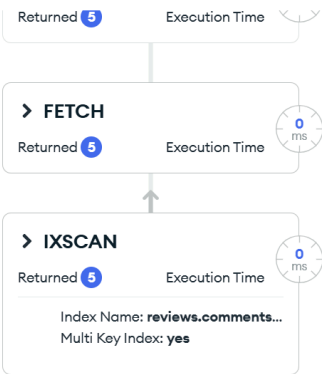
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Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Visual Tree Raw Output

+
-



Query Performance Summary

- 5 documents returned
- 5 documents examined
- 0 ms execution time
- Is not sorted in memory
- 5 index keys examined

Query used the following index:

_fts (text)
_ftsx ↑

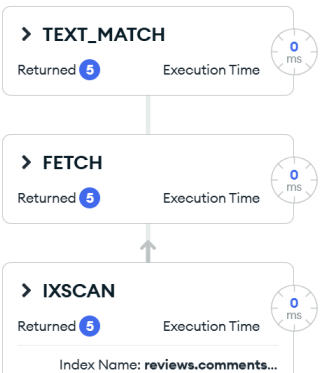
Close

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Visual Tree Raw Output

+
-



Query Performance Summary

- 5 documents returned
- 5 documents examined
- 0 ms execution time
- Is not sorted in memory
- 5 index keys examined

Query used the following index:

_fts (text)
_ftsx ↑

Close

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Interpret ⓘ

Visual Tree Raw Output

> COLLSCAN

Returned 0 Execution Time



Documents Examined: 5

+
-

Query Performance Summary

- 0 documents returned
- 5 documents examined
- 0 ms execution time
- Is not sorted in memory
- 0 index keys examined
- No index available for this query.

Close